MECH 151: Technical Design — Prototyping

Unit Competency & Knowledge Assessment Questions

Drafted from the Unit Reference Slide Deck and syllabus schedule — 5 questions per unit, answer key included

Each unit below carries five questions blending recall (multiple choice / identification), explanation (short answer), and applied reasoning (scenario, calculation, or design judgment), matching the three unit competency checks on the syllabus (Weeks 8, 11, 16). A single answer key with brief rationale for every question, grouped by unit, appears at the end of this document.

Unit 1 — CAD Fundamentals (Basic)

Weeks 4–8. Assessment: Unit 1 Competency Check, Week 8.

Question 1 — *Multiple Choice*

Which sequence correctly describes the dimensional progression used to build a solid parametric model?

A. 3D Solid → 2D Profile → 1D Line → 0D Point

B. 0D Point → 1D Line/Curve → 2D Polygon/Profile → 3D Solid/Volume

C. 1D Line → 0D Point → 3D Solid → 2D Profile

D. 2D Profile → 3D Solid → 0D Point → 1D Line

Question 2 — *Short Answer — Constraint States*

A sketch still has some geometry that can move freely when dragged, even though several dimensions have been added. Is this sketch under-defined, fully defined, or over-defined? What general rule should be followed before adding dimensions, and why?

Question 3 — *Applied — Orthographic Projection*

In third-angle projection (the U.S. standard), where is the top view placed relative to the front view, and where is the right-side view placed? A student places the top view below the front view on a drawing — is this correct convention for a U.S. drawing, and what would you tell them?

Question 4 — *Applied — Measurable Properties*

Explain the difference between a 2D profile's centroid and a 3D solid's center of gravity. Then explain why assigning an incorrect material density in the CAD software produces a wrong mass moment of inertia even when the model's geometry is completely accurate.

Question 5 — *Short Answer — Design Philosophy*

Two students model the same bracket using completely different feature trees, yet both receive full credit. What criteria actually determine whether a parametric model is judged “correct” in this course, given that there is no single right feature tree for a given shape?

Unit 2 — Design for Assembly (DFA)

Weeks 8–11. Assessment: Unit 2 Competency Check, Week 11.

Question 1 — *Short Answer — Mates & DOF*

A free rigid body has 6 degrees of freedom. Explain how a concentric mate plus a coincident mate, applied together to two cylindrical parts being assembled, reduce those degrees of freedom.

Question 2 — *Applied — Joining Method Selection*

You are designing a 3D-printed plastic enclosure that must be assembled and disassembled multiple times for maintenance, using threaded fasteners into printed bosses. What minimum thread engagement depth should you plan for relative to the screw diameter, and why might a heat-set threaded insert be preferred over threading directly into the printed plastic?

Question 3 — *Applied — Joining Method Selection*

A design needs a joint that flexes repeatedly to allow opening and closing, without a separate hinge pin or hardware. Which joining approach fits this need, what is this joint type called, and which two plastics are specifically noted as tolerating repeated flexing?

Question 4 — *Short Answer — DFA / BOM*

Using a housing / cover plate / gasket / fastener BOM as a reference, explain why both “reducing part count” and “preferring off-the-shelf components” are DFA principles that reduce assembly cost.

Question 5 — *Applied — Sourcing*

Why does importing a fastener model from an online repository like McMaster-Carr (with a real part number) strengthen a bill of materials, compared to modeling a generic bolt from scratch? What limitation do imported models typically have once brought into your assembly?

Unit 3 — Design for Manufacturing (DFM)

Weeks 12–16. Assessment: Unit 3 Competency Check, Week 16.

Question 1 — *Short Answer — GD&T Families*

Sort the following GD&T characteristics into their correct family — Form, Orientation, Location, Profile, or Runout — and note which families require a referenced datum: flatness, perpendicularity, position, profile of a surface, circular runout.

Question 2 — *Applied — Feature Control Frame*

Decode the feature control frame "⌀ 0.2 Ⓜ A B C." What does each element specify, and in what order do the three datums establish the datum reference frame?

Question 3 — *Applied — Modifiers*

Explain the difference between the MMC and LMC modifiers, and why a hole drilled larger than its worst-case (MMC) size is allowed extra ("bonus") positional tolerance.

Question 4 — *Applied — Process & Defects*

A printed part comes off the printer with its edges curling upward and lifting away from the bed. Name this defect, its cause, and at least one fix. Then explain why this same design concern (cooling-driven distortion) would not apply if the part were laser-cut instead.

Question 5 — *Applied — Safety & Materials*

Why must PVC (or other chlorinated plastics) never be cut on a laser cutter? Also name one PPE requirement for operating rotating shop equipment such as a drill press or lathe.

Answer Key

Unit 1 — CAD Fundamentals (Basic)

Q1.

A. 3D Solid → 2D Profile → 1D Line → 0D Point

B. 0D Point → 1D Line/Curve → 2D Polygon/Profile → 3D Solid/Volume

C. 1D Line → 0D Point → 3D Solid → 2D Profile

D. 2D Profile → 3D Solid → 0D Point → 1D Line

B — a sketch builds up from points, to lines/curves, to a closed 2D profile, which is then given a third dimension (via extrude, revolve, sweep, or loft) to become a 3D solid.

Q2.

Under-defined — geometry that can still move has not had its position or size fully pinned down by relations and dimensions. The rule is to add geometric relations (coincident, parallel, tangent, concentric, symmetric, etc.) before adding dimensions, because relations capture the design's intended behavior (e.g., “always centered,” “always tangent”) more robustly than dimension values alone, and reduce the number of dimensions needed to fully define the sketch.

Q3.

In third-angle projection, the top view goes above the front view, and the right-side view goes to the right of the front view. Placing the top view below the front view follows first-angle projection (the convention used in much of the rest of the world, not the U.S. standard), so the student's layout would be non-conforming for a U.S./ASME Y14.5 drawing — they should move the top view above the front view (or explicitly label the drawing as first-angle if that convention was intentional).

Q4.

A centroid is the geometric center of a 2D area (independent of material); center of gravity is the mass-weighted balance point of a 3D solid, which depends on both geometry and how mass (density × volume) is distributed through it. Because mass moment of inertia is computed from the mass distribution, an incorrect density scales or skews the calculated mass at every point in the solid, so the result is wrong even though the geometry itself is perfect — this is the “garbage in, garbage out” problem: correct geometry cannot compensate for incorrect material properties or a non-watertight solid.

Q5.

Models are judged on: (1) whether they capture the design intent (the reasoning behind the shape, via relations, dimension choices, and feature order/references), (2) whether they rebuild cleanly without errors, and (3) whether they remain easily editable. A model is not judged by matching one expected sequence of features — effectively infinite valid feature trees can produce the same shape.

Unit 2 — Design for Assembly (DFA)

Q1.

A concentric mate aligns the two parts' cylindrical axes, removing the two translational degrees of freedom perpendicular to that axis (and constraining rotation about the two axes perpendicular to it), leaving the parts free to slide along and rotate about the shared axis. A coincident mate (e.g., mating a face of one part flush to a face of the other) then removes the remaining translational degree of freedom along that axis. Together, the two mates can reduce the pair from 6 DOF down to a single rotational degree of freedom about the shared axis (or fully fix it, if rotation is also constrained by another feature).

Q2.

Plan for roughly 1.5–2× the screw diameter as the minimum thread engagement depth. A heat-set insert (a brass insert pressed in with a soldering iron) is often preferred over threading directly into printed plastic because printed plastic threads wear out and strip after repeated assembly/disassembly cycles, whereas a metal insert provides a durable, reusable thread that tolerates repeated maintenance cycles without degrading the boss.

Q3.

A compliant mechanism / living hinge is the appropriate approach — a thin, flexible section of material molded or printed as one piece with the surrounding part, allowing repeated flexing without added hardware. PP (polypropylene) and PE (polyethylene) are the two materials specifically noted as tolerating repeated flexing without fatigue-cracking.

Q4.

Reducing part count means fewer items to source, inspect, handle, and mate during assembly — each additional unique part adds handling time, potential for error, and inventory/BOM complexity. Preferring off-the-shelf components (like a standard socket head cap screw, HW-0425) avoids the cost and lead time of custom-manufacturing a part that a catalog item already satisfies, and standard hardware is typically cheaper, better documented, and easier to replace than a custom equivalent.

Q5.

An imported model with a real part number strengthens the BOM because it ties directly to a purchasable, verifiable catalog item — anyone building the assembly can order the exact part, rather than guessing at dimensions from a hand-modeled generic bolt that may not match anything actually manufactured. The limitation is that imported models are typically “dumb” solids: they carry no feature history/parametric tree, so they can be positioned and mated but not easily edited or re-parameterized within your CAD system.

Unit 3 — Design for Manufacturing (DFM)

Q1.

Flatness — Form (no datum required). Perpendicularity — Orientation (requires a datum). Position — Location (requires datum(s); position is the most commonly used location control). Profile of a surface — Profile (can be datum-related or not, depending on the callout). Circular runout — Runout (requires a datum axis).

Q2.

⌀ 0.2 = a cylindrical (diametral) tolerance zone of 0.2, meaning the controlled feature's axis must fall within a 0.2-diameter cylinder. Ⓜ = the modifier for maximum material condition (MMC), meaning the stated tolerance applies when the feature is at its worst-case (most material) size, with bonus tolerance available as the feature departs from MMC. A B C = the datum reference frame, applied in priority order: A is the primary datum (constrains the most degrees of freedom, typically contacted first), B is secondary, and C is tertiary — together the three progressively lock all 6 degrees of freedom of the part relative to the inspection/measurement setup.

Q3.

MMC (maximum material condition) is the size at which a feature contains the most material — for a hole, its smallest allowed diameter; for a shaft, its largest. LMC (least material condition) is the opposite — the least amount of material a feature can have. Bonus tolerance is granted because the total functional requirement is that a mating part must still fit/assemble; if a hole is drilled larger than its MMC (smallest) size, there is more clearance available for that mating part, so the hole's position is allowed to drift farther off-target while assembly is still guaranteed — the bonus tolerance exactly equals how far the actual size has departed from MMC.

Q4.

This is warping — caused by the part's edges and corners cooling and shrinking faster than the center of the part, causing the corners to curl. Fixes include using a brim or raft, printing in an enclosure to slow/even out cooling, and using a heated bed. This concern doesn't apply to laser cutting because laser cutting is a subtractive, sheet-based process performed on flat stock at essentially uniform temperature throughout the cut — there is no layer-by-layer thermal deposition or differential cooling of built-up material, so warping of this kind isn't a relevant failure mode.

Q5.

Cutting PVC or other chlorinated plastics on a laser cutter releases chlorine gas, which is toxic to breathe and corrosive to the machine's optics and hardware — these materials must never be laser cut. For rotating equipment (drill press, lathe, etc.), required PPE/practice includes ANSI Z87.1 safety glasses and closed-toe shoes; gloves must not be worn near a rotating spindle, chuck, or bit, because a caught glove can pull the hand into the machine.